

CRYPTOFLOW AI

Crypto Flow & Fraud Analytics Platform

Project Concept & Functional Documentation

1. Project Overview

CryptoFlow AI is a cybersecurity and financial-forensics platform designed to analyze cryptocurrency transaction flows and identify suspicious behavioral patterns. The platform does not attempt to decide that a person is a criminal. Instead, it provides investigators with evidence-based risk indicators, transaction graphs, wallet analytics, alerts, and AI-generated summaries that help them investigate potentially fraudulent activity.

The core idea is simple: a blockchain records cryptocurrency transactions, but a large number of interconnected transactions can be difficult for a human investigator to interpret. CryptoFlow AI converts this raw transaction history into an understandable wallet-to-wallet flow graph, extracts behavioral features, detects anomalies and suspicious patterns, assigns an explainable risk score, and presents the findings through an investigation dashboard.

2. Problem Statement

Cryptocurrency transactions are recorded digitally and can move rapidly across multiple wallet addresses. When fraud or scams occur, funds may pass through several wallets, split across many addresses, be consolidated into another wallet, or move through complex transaction paths. Although blockchain transaction records can be publicly available, manually following thousands of transactions and identifying meaningful suspicious patterns is difficult and time-consuming.

Investigators and compliance teams need a system that can answer questions such as:

- Where did the cryptocurrency originate?
- Where did the funds move after a suspicious transaction?
- Which wallets are connected to the investigated wallet?
- Are there unusual transaction patterns or rapid movements of funds?
- Which wallets or transactions deserve further investigation?
- What evidence caused a wallet or transaction to receive a high risk score?

3. Proposed Solution

CryptoFlow AI proposes an integrated analytics platform that ingests blockchain transaction data, normalizes it, constructs a transaction graph, extracts wallet and transaction features, detects suspicious patterns using explainable rules and optional machine-learning models, calculates a risk score, and generates an AI-assisted investigation summary.

The platform is intended as an investigation-support system rather than an autonomous criminal judgment system. A high-risk score means that the observed activity contains indicators that warrant further investigation; it does not prove that fraud has occurred.

4. Core Concept: Crypto Flow

Crypto flow means tracing the movement of cryptocurrency between blockchain addresses. For example, a suspicious flow may conceptually look like:

Victim → Wallet A → Wallet B → Wallet C → Exchange

In the platform's graph model, each wallet/address is represented as a node and each relevant transaction is represented as a directed edge. The edge can carry information such as amount, timestamp, and transaction hash.

5. Main Website Flow

1. **Step 1 — User Input:** The investigator enters a supported wallet address or transaction hash and selects the blockchain/network.
2. **Step 2 — Data Ingestion:** The backend requests relevant blockchain transaction data from a blockchain data provider/API or other approved data source.
3. **Step 3 — Data Processing:** Raw transactions are validated, normalized, deduplicated, timestamp-normalized, and converted into a consistent internal format.
4. **Step 4 — Transaction Graph:** The platform builds wallet-to-wallet relationships. Wallets become nodes and transactions become directed edges.
5. **Step 5 — Feature Extraction:** The system calculates transaction frequency, amounts, wallet connectivity, transaction velocity, incoming/outgoing ratios, fan-in/fan-out behavior, and other analytical features.
6. **Step 6 — Fraud/Anomaly Detection:** Rules and, where suitable, ML models analyze the features to identify suspicious patterns and anomalies.
7. **Step 7 — Risk Scoring:** Evidence from the detection layer is combined into an explainable 0–100 risk score and a risk category.
8. **Step 8 — AI Investigation Summary:** An LLM receives structured findings and produces a human-readable explanation of the observed indicators.
9. **Step 9 — Dashboard:** The investigator sees the transaction graph, wallet profile, alerts, risk indicators, charts, and AI summary.
10. **Step 10 — Report:** The platform can generate a case/investigation report containing transaction evidence, risk factors, flow paths, and summary.

6. Main Features

1. Wallet & Transaction Analyzer

Analyze a wallet address or transaction hash and display transaction count, incoming/outgoing activity, timestamps, amounts, connected addresses, and other available metadata.

2. Crypto Flow Visualization

Display the movement of funds as an interactive network graph. Investigators can follow wallet-to-wallet paths, click nodes, inspect transactions, and expand connected wallets.

3. Fraud & Anomaly Detection

Detect suspicious behavioral indicators such as rapid fund movement, abnormal transaction frequency, fan-out, fan-in, circular flows, unusual amounts, and other configurable patterns.

4. Explainable Risk Scoring

Generate a 0–100 risk score and show the factors contributing to it. The platform should explain the score instead of presenting an unexplained black-box prediction.

5. Wallet Risk Profile

Provide a detailed profile for an investigated wallet, including activity statistics, connectivity, transaction behavior, detected indicators, and current risk level.

6. AI Investigation Assistant

Convert structured analytics into an investigator-friendly summary, explain why a wallet was flagged, highlight important paths, and suggest which connected entities deserve review.

7. Alerts & Case Management

Show high-risk wallets or transactions in an alert queue and allow investigators to open, review, and track cases.

8. Investigation Report

Generate a downloadable report containing wallet/transaction details, suspicious patterns, flow information, risk score, and an AI-generated summary.

7. Suspicious Patterns to Detect

- **Rapid Fund Movement:** Funds move through several addresses in a short time window.
- **Fan-Out:** One address distributes funds to many addresses, especially when the behavior is unusually concentrated in time.
- **Fan-In:** Many addresses send funds into one address, which may be relevant for collection or consolidation analysis.
- **Circular Flow:** Funds move through a connected path and eventually return to an earlier address or connected cluster.
- **Abnormal Transaction Velocity:** A wallet suddenly performs many transactions compared with its normal or peer behavior.
- **Unusual Transaction Amounts:** Transaction values significantly deviate from the wallet's observed historical behavior.
- **Suspicious Connections:** A wallet interacts with addresses or entities already marked as high-risk by a trusted intelligence source, where such data is available.

Important: these are risk indicators, not proof of criminal activity. Legitimate exchanges, businesses, services, and high-volume users can also create complex transaction patterns.

8. Risk Scoring Model

A practical MVP can begin with an explainable weighted scoring system. For example, each detected indicator contributes points to a normalized 0–100 score. Later, a machine-learning model can be added if the team obtains a sufficiently reliable labeled dataset.

Risk Score	Category	Meaning
0–30	Low	Few or weak risk indicators
31–60	Medium	Some indicators require review
61–80	High	Multiple significant indicators
81–100	Critical	Strong combination of risk indicators; prioritize investigation

9. Technical Architecture

Recommended architecture for a B.Tech implementation:

- Frontend: React
- Backend/API: Node.js + Express
- Analytics service: Python
- Database: MongoDB or PostgreSQL
- Graph analysis: NetworkX; optionally Neo4j for larger graph workloads
- Graph visualization: React Flow, D3.js, or Cytoscape.js
- ML: scikit-learn / XGBoost for supervised models when suitable labeled data is available; Isolation Forest or similar methods for anomaly detection
- AI explanation: Gemini or another LLM
- Deployment: Vercel for frontend and Render/Railway/AWS/etc. for backend and analytics service

10. Technical Data Flow

The end-to-end technical pipeline can be represented as:

Blockchain Data → Data Ingestion → Normalization → Transaction Graph → Feature Extraction → Rule/ML Detection → Risk Scoring → AI Explanation → Dashboard → Report

11. Example Investigation

Suppose a victim reports that cryptocurrency was sent to Wallet A. The investigator enters Wallet A into CryptoFlow AI. The system retrieves relevant transaction data, identifies that Wallet A forwarded most of the received value to several newly active addresses within a short period, and detects a fan-out pattern followed by consolidation. The platform visualizes the path, calculates a high risk score based on the observed indicators, and generates an investigation summary.

Example output:

- Risk Score: 87/100 — High Risk
- Rapid fund movement detected
- High fan-out activity detected
- Multiple newly active connected addresses
- Two possible circular paths detected
- Recommended action: investigate connected wallets and relevant transaction hashes

12. Suggested Website Pages

Dashboard: Overall transaction volume, suspicious wallets, high-risk alerts, and analytics.

Analyze: Input wallet/transaction hash and start an investigation.

Flow Explorer: Interactive transaction graph and wallet-to-wallet flow.

Wallet Profile: Detailed activity, connections, risk indicators, and score.

Alerts: List of high-risk transactions/wallets requiring review.

Analytics: Charts showing transaction behavior, risk distribution, and suspicious patterns.

AI Investigation: Natural-language explanation of detected indicators.

Reports: Case reports and downloadable investigation summaries.

13. What Makes the Project Different

- It focuses on the movement of funds, not simply cryptocurrency prices.
- It combines graph-based transaction tracing with fraud analytics.
- Risk scores are explainable through visible evidence and indicators.
- AI is used as an investigation assistant rather than as an unsupported black-box fraud judge.
- The same transaction graph can support both visual investigation and analytical feature extraction.
- The system can be demonstrated using public/test blockchain data without handling real user funds.

14. MVP Scope for a B.Tech/Hackathon Team

To keep the project achievable, the first version should support one blockchain/network and a limited set of well-defined fraud indicators.

11. Support one blockchain first.
12. Analyze wallet address and transaction hash.
13. Fetch and store transaction history.
14. Build an interactive transaction graph.
15. Implement 3–5 explainable suspicious-pattern rules.
16. Calculate a transparent risk score.
17. Add an AI explanation layer.
18. Create a clean investigator dashboard.
19. Add PDF/JSON investigation export if time permits.

15. One-Line Pitch for Judges

“CryptoFlow AI traces cryptocurrency movement across connected wallets, detects suspicious transaction patterns, assigns explainable risk scores, and gives investigators an AI-assisted view of where the money moved and why the activity deserves attention.”

16. Important Design Principle

The platform should never claim that a wallet owner is a criminal solely because an algorithm gives a high score. A high score should mean that the transaction behavior contains indicators that warrant further investigation. This distinction makes the system more responsible, explainable, and technically defensible.